Visual Dimension Estimation of Bolts and Washers in Oblique-View Industrial Tabletop Scenarios

Qianxi Liu, Junfeng Cui, Yitao Pan, Yongsheng Ou, and Zhangxi Xiong

***Abstract*—In embodied intelligent industrial assembly, the geometric parameters of metallic fasteners such as bolts and washers—including length, diameter, and inner/outer diameter—are critical for assembly matching and quality consistency. Conventional RGB-D point-cloud methods are susceptible to specular reflection, causing depth voids and noise that destabilize two-point distance measurements. To address these challenges, this paper proposes a geometric dimension estimation framework based on RGB-D imagery: to handle perspective distortion under oblique viewing, we establish a unified planar metric reference by estimating a homography matrix through square- or round-tabletop coordinate systems, performing perspective rectification and computing physical dimensions based on oriented bounding boxes (OBBs) from a rotated object detector. Our contributions include: (1) a two-stage refinement strategy with rotation-aligned regions of interest (ROIs) for robust inner-hole detection of washers; (2) a height-prior and imaging-geometry compensation mechanism that corrects the systematic projection bias caused by bolts being elevated above the reference plane; and (3) a RANSAC-based robust dimension estimation method that suppresses outliers in both tabletop calibration and size computation. Furthermore, we construct a high-quality industrial fastener dataset comprising over 10,000 images, covering both square and round tabletop scenarios, for systematic evaluation of dimension estimation and localization errors. Experimental results demonstrate that the proposed method achieves millimeter-level accuracy across different scenarios and viewpoints, maintaining high consistency within a 5 mm tolerance, and significantly outperforms methods without geometric compensation and robust optimization.**

***Index Terms*—Industrial metrology, machine vision measurement, homography rectification, RGB-D, error compensation.**

I. Introduction

WITH the advancement of embodied intelligence and smart manufacturing, industrial visual perception is expanding from "object detection and grasp localization" toward "geometric estimation and process closed-loop control" [21], [58]. In assembly, sorting, and online quality inspection, the key geometric parameters of standard fasteners—bolt total length and diameter, washer outer and inner diameter—directly affect assembly clearance, joint stiffness, and reliability [25], [35], [48]. Recent vision-guided robotic assembly systems further demonstrate the practical need for measurement-grade geometric outputs integrated into the manipulation loop [6], [23].

Traditional non-contact measurement typically relies on RGB-D reconstruction of point clouds and computes dimensions through endpoints, axes, or local feature-point distances. For example, bolt-loosening quantification based on exposed-length changes fundamentally depends on the continuous availability of point-cloud endpoints and contours [25]. However, in real industrial environments, specular reflection, highlight saturation, and boundary aliasing of metallic parts severely degrade the depth reliability of consumer-grade RGB-D sensors at boundaries, causing voids, discontinuities, and outliers. These artifacts lead to broken contours and unavailable endpoints, rendering direct two-point distance measurement from sparse point-cloud samples unreliable [5], [30], [42], [43]. Recent surveys and benchmarks confirm that specular and transparent surfaces remain primary failure modes for low-cost depth sensing and depth enhancement [37], [38]. Approaches for such non-Lambertian surfaces leverage synthetic augmentation, domain randomization, and depth restoration to mitigate sensor failure, yet consistent and repeatable boundary recovery under complex conditions remains a bottleneck [5], [9], [26], [42]. Moreover, sparse-depth-guided completion and propagation methods improve overall depth quality, but boundary errors and structural discontinuities still commonly dominate the systematic bias in downstream geometric estimation [19], [33], [36], [46], [47]. Diffusion-based depth completion has recently shown promise on specular surfaces, but the computational cost often limits real-time deployment in industrial pipelines [31], [49]. To illustrate these failure modes concretely, Fig. 1 shows an example of depth voids under metallic specular conditions: depth is clearly missing and structurally broken in the strong-reflection regions, directly compromising the continuity of point-cloud endpoints and contours and thus destabilizing downstream geometric estimation.

Beyond depth failure, static tabletop measurement commonly employs oblique camera deployment to balance field of view and safety distance. Oblique perspective causes pixel scale to vary with position; measuring directly in the pixel domain or on unstable point clouds introduces systematic bias that precludes millimeter-level accuracy. Planar homography provides a unified mapping from the image domain to a planar metric domain: by establishing correspondences on a reference plane, perspective distortion can be rectified into a coordinate system with consistent scale, enabling metric measurement in the rectified domain [16], [34], [41], [55]. While learning-

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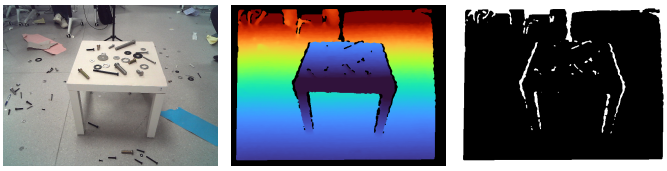

(a) RGB image (b) Depth image (c) Hole mask

Fig. 1. Depth void example under metallic specular reflection. Strong highlights cause widespread depth loss and structural discontinuities, rendering point-cloud-based endpoint measurement unreliable.

based homography estimation has progressed rapidly [7], [24], [53], in constrained industrial settings, classical geometry-driven estimation remains attractive due to interpretability and deterministic error behavior. In practice, tabletop segmentation boundaries often contain gaps and outliers, making homography estimation noise-sensitive; robust model fitting (e.g., RANSAC) and edge/corner operators are thus crucial to improve repeatability [1], [11], [15]. For round tabletops that lack corners, the oblique projection of a circle appears as an ellipse; conic constraints can be exploited for calibration and rectification, but ellipse fitting is sensitive to outliers and must be combined with robust fitting strategies [2], [12], [20].

On the "detection-to-metrology" pipeline, deep-learning-based detection and instance segmentation significantly improve the detectability of industrial parts under complex backgrounds [17], [18], [28], [29], [39], [40]. However, detection outputs typically serve localization rather than metrology: box width and height do not directly equate to real dimensions, especially when targets are rotated and foreshortened. Oriented bounding box (OBB) detection explicitly models targets as rotated rectangles, improving localization consistency in rotated scenarios [8], [14], [50]–[52]. The YOLO family has further enabled practical OBB deployment with native rotated-box support [22]. Recent advances on continuous OBB representations mitigate angle discontinuity issues in regression, and weakly supervised OBB learning reduces annotation cost [27], [32], [54], [57]. Nevertheless, achieving industrial metrology accuracy still requires further "geometrizing" detection outputs into keypoints/structural elements and computing dimensions in a scale-consistent metric coordinate system while explicitly handling systematic errors introduced by perspective and non-coplanarity. Keypoint-based pose estimation in industrial settings shows that geometric priors can significantly enhance measurement precision [3], [13], [56].

Based on these observations, this paper addresses the following task: under oblique-view deployment, given RGB or RGB-D input, perform instance-level detection, dimension estimation, and planar coordinate localization of bolts and washers on a tabletop, outputting metric coordinates referenced to a tabletop coordinate system along with key geometric dimensions. The core idea is to unify all measurements in a tabletop metric domain: first, a stable tabletop reference is used to estimate homography and rectify perspective distortion; then, leveraging OBB detections and robust geometric fitting, key dimensions are computed in the rectified metric domain while compensating systematic projection bias caused by bolt height, thereby maintaining consistent dimension and position outputs even when depth information is incomplete or locally invalid (e.g., the specular voids shown in Fig. 1). Experiments demonstrate millimeter-level accuracy across different scenarios and viewpoints, maintaining high consistency within a 5 mm tolerance.

The main contributions are summarized as follows:

1) We propose a detection-to-metrology integrated framework that unifies dimension estimation and planar localization in a tabletop metric domain.
2) We introduce robust tabletop rectification pipelines for both square and round tables, combining structural regularization with RANSAC-based outlier suppression for repeatable homography estimation.
3) We present an interpretable imaging-geometry compensation mechanism for height-induced projection bias of bolts, improving cross-position consistency under oblique viewing.

II. Related Work

Industrial visual metrology in tabletop scenarios typically faces three key challenges: (i) establishing a stable and repeatable planar metric coordinate system under oblique viewing; (ii) obtaining reliable geometric constraints when specular metallic surfaces cause depth voids and boundary discontinuities; and (iii) converting detection outputs into metrologically meaningful structural elements while suppressing systematic errors from rotation, perspective, and non-coplanarity. Corresponding to our "tabletop reference construction—detection-to-metrology—error compensation" pipeline, relevant literature spans the following directions.

A. Planar Metric Rectification and Homography

Planar homography is a core tool connecting the image domain to planar geometry. Classical calibration and rectification methods provide interpretable mappings and support metric measurement under planar assumptions [16], [34], [41], [55]. In engineering scenarios, homography stability strongly depends on robust planar structural constraints; when segmentation boundaries contain gaps and outliers, robust estimation and edge/corner operators become essential for repeatability [1], [11], [15]. Learning-based homography estimation has been explored for generic image alignment and dynamic scenes [7], [24], [53], but deterministic geometry-driven methods remain preferable in constrained metrology settings due to controllable error behavior.

B. Ellipse and Circle Constraints under Perspective

When the tabletop lacks explicit corner points (e.g., round tables), circles and ellipses provide alternative conic constraints: a planar circle appears as an ellipse under perspective imaging and can be used to infer metric rectification. However, ellipse detection and parameter estimation are sensitive to contour quality, requiring robust fitting combined with direct least-squares strategies to suppress outliers [12], [20]. Conic-based calibration from circular objects has been studied as a practical calibration primitive [2]. Our round-table pipeline

combines robust ellipse regularization with depth-guided plane recovery and circle fitting to yield repeatable homography estimation under boundary noise.

C. RGB-D Reliability on Specular and Transparent Surfaces

Depth voids and outliers at specular boundaries dominate errors in RGB-D metrology [30], [43]. For transparent and highly reflective objects, approaches combining synthetic data, domain randomization, and depth restoration improve depth availability [5], [9], [26], [42]. Surveys and benchmarks highlight persistent difficulty on non-Lambertian surfaces [37], [38]. Propagation-based and sparse-depth-guided completion methods improve global depth quality but remain challenged by boundary consistency, which often dominates systematic downstream measurement bias [19], [33], [36], [46], [47]. Diffusion-based depth completion offers a promising direction for specular scenes but is computationally expensive for real-time industrial deployment [31], [49].

D. From Detection to Metrology: OBB and Geometry-Aware Measurement

General object detection and instance segmentation provide strong support for the detectability of industrial parts [17], [18], [28], [29], [39], [40]. For rotated objects, OBB detection has developed a systematic landscape in datasets, feature alignment, and angle parameterization [8], [14], [50]–[52]. Practical OBB deployment has also been enabled by modern one-stage detectors with native rotated-box support [22]. Recent advances address angle discontinuity via continuous orientation/OBB representations and reduce labeling cost via weak supervision [27], [32], [54], [57]. However, metrology tasks require not only correct localization but also scale consistency, structural interpretability, and explicit error modeling. Thus, converting OBB outputs into metric keypoints and performing robust geometry in a rectified metric domain is often necessary, especially under non-coplanarity and perspective distortions [3], [13], [56].

E. Fastener Visual Inspection and Measurement

Fastener-related research includes looseness monitoring, condition assessment, defect detection, and geometric estimation. Bolt looseness monitoring demonstrates that imaging conditions and structural characteristics strongly affect estimation stability [35]. Point-cloud-based geometric estimation leverages 3D information but depends on depth continuity [25]. Surveys on vision-based bolt assessment show that most methods focus on detection and classification rather than precise dimension measurement [48]. These studies indicate that "visual interference + insufficient geometric constraints + non-negligible systematic errors" often coexist in industrial scenes, motivating frameworks that unify robust planar references, geometry-aware measurement, and interpretable error compensation.

F. Deep Learning in Optical Metrology

Deep learning has been increasingly applied to optical metrology and industrial inspection, including error compensation and measurement-oriented perception. Reviews summarize methodological trends and emphasize that combining learning with explicit physical/geometry priors is often critical under strict accuracy constraints [10], [44]. For metallic objects, learning-based specular removal and enhancement have also been explored to improve downstream measurement robustness [4].

III. Method

A. Problem Definition and System Overview

Given an RGB image I captured under oblique-view deployment, an aligned depth map/point cloud D, and the camera intrinsic matrix K, our goal is to estimate millimeter-level key geometric parameters for the set of tabletop fasteners $\mathcal{O} = \{\text{bolt}, \text{washer}\}$: bolt length L and diameter d, washer outer diameter d_{out} and inner diameter d_{in}, as well as the 2D metric coordinates of each instance in the tabletop coordinate system.

As discussed in the introduction, depth voids caused by metallic specular reflections make point-cloud-endpoint-based direct distance measurement difficult to stabilize. Therefore, we unify all geometric computations in the *tabletop metric domain*. The core idea is as follows: we establish a homography mapping from the image to a metric canvas using the tabletop planar structure, converting perspective distortion into a rectifiable planar warping problem; in the metric domain, we perform robust geometric fitting and dimension computation under boundary constraints from RGB masks; for bolts that violate the coplanarity assumption, we introduce an imaging-geometry-based height compensation to eliminate systematic bias. The overall pipeline is shown in Fig. 2.

Let the homogeneous image coordinate be $\mathbf{p} = [u, v, 1]^\top$, and the homogeneous coordinate on the metric canvas be $\mathbf{q} = [x, y, 1]^\top$. We estimate a homography matrix $H_{\text{img}\to\text{can}}$ from the image to the metric canvas such that

$$\mathbf{q} \sim H_{\text{img}\to\text{can}}\mathbf{p}, \qquad (x, y) = \left(\frac{q_1}{q_3}, \frac{q_2}{q_3}\right). \tag{1}$$

The metric canvas has a fixed size of $N \times N$ pixels. We define the millimeter conversion factor as

$$s_{\text{mm/px}} = \frac{W_{\text{mm}}}{N}, \tag{2}$$

where W_{mm} denotes the true tabletop extent (side length for a square table and diameter for a round table, in mm). All dimension estimation in this paper follows the same principle: **keypoints are first projected to the metric domain via $H_{\text{img}\to\text{can}}$, then Euclidean distances are computed in the metric domain and multiplied by $s_{\text{mm/px}}$**, rather than measuring directly in the original pixel domain.

Based on the above metric framework, the system consists of the following five sequential modules: (i) **Tabletop segmentation and metric rectification**—recover the tabletop geometric model from the tabletop mask and estimate $H_{\text{img}\to\text{can}}$ and $s_{\text{mm/px}}$; (ii) **Part detection and ROI refinement**—output

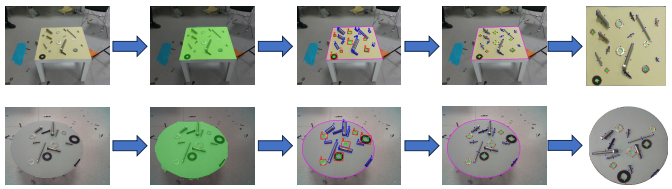

Fig. 2. Overall pipeline. The system consists of five modules: tabletop metric rectification, part detection and ROI refinement, keypoint construction and homography projection, metric-domain dimension computation, and height compensation.

rotated bounding boxes (OBBs) for bolts and washers, and perform a secondary detection for washer inner holes when needed; (iii) **Keypoint construction and metric-domain projection**—extract keypoints from OBBs and project them to the metric domain via homography; (iv) **Metric-domain dimension computation**—estimate bolt and washer dimensions in the metric domain; and (v) **Height compensation**—geometrically correct the systematic perspective-projection bias induced by bolt non-coplanarity.

B. Tabletop Metric Rectification

The objective of metric rectification is to recover the tabletop geometric model from the tabletop mask M (and optional depth D) and then solve $(H_{\text{img}\to\text{can}},\, s_{\text{mm/px}})$. We design separate rectification strategies for two typical tabletop shapes, i.e., square tables and round tables. The difference lies only in the estimation of tabletop geometric parameters and the construction of the canonical metric canvas.

1) Square Table: Quadrilateral Regularization and Homography Estimation: A square table appears as a quadrilateral under oblique viewing. However, the boundary of the segmentation mask M typically contains burrs, gaps, and local outliers, making direct corner extraction unstable. Therefore, we regularize the contour into a quadrilateral model and obtain refined corners via robust line fitting, followed by solving a homography mapping to a canonical square canvas. The pipeline is illustrated in Fig. 3.

a) Multi-scale quadrilateral approximation: Let $\mathcal{C} = \{\mathbf{c}_i\}$ be the largest external contour point set extracted from the tabletop mask M. Using a single approximation tolerance ϵ suffers from a trade-off: a large ϵ shifts corner locations, while a small ϵ retains noisy vertices and fails to converge to a quadrilateral. We adopt a multi-scale strategy by iteratively applying contour approximation over a set of decreasing tolerances $\epsilon \in [\epsilon_{\max}, \epsilon_{\min}]$, and selecting the first result that yields exactly four vertices as the candidate quadrilateral $\{\mathbf{p}_k\}_{k=1}^4$. This strategy makes corner extraction less sensitive to tolerance selection and remains stable under segmentation quality variation.

b) RANSAC-based line refinement: The candidate quadrilateral provides only coarse corner locations. To further improve accuracy, we assign contour points to four subsets $\{\mathcal{G}_k\}_{k=1}^4$ according to their minimum distance to the four candidate edges, and perform RANSAC line fitting on each subset to obtain ℓ_k. Introducing RANSAC makes edge fitting

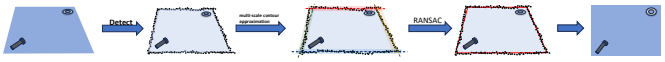

Fig. 3. Square-table metric rectification pipeline. Starting from the mask contour, multi-scale approximation, RANSAC-based line refinement, and line-intersection corner estimation yield robust quadrilateral corners, which are then mapped to a canonical square canvas.

robust to local gaps and outliers [11]. The refined corners are computed as the intersections of adjacent lines:

$$\tilde{\mathbf{p}}_k = \ell_k \cap \ell_{k+1}, \quad k = 1, \ldots, 4 \tag{3}$$

c) Homography estimation: We sort $\{\tilde{\mathbf{p}}_k\}$ into $\{\text{TL}, \text{TR}, \text{BR}, \text{BL}\}$ order, and define the four corners of the canonical canvas as

$$\mathbf{q}_1 = (0,0),\ \mathbf{q}_2 = (N,0),\ \mathbf{q}_3 = (N,N),\ \mathbf{q}_4 = (0,N). \tag{4}$$

The homography $H_{\text{img}\to\text{can}}$ is solved from these four point correspondences. The millimeter conversion factor is given by Eq. (2), where W_{mm} is the true side length of the square tabletop.

2) Round Table: Contour–Plane–Circle Joint Calibration: A round table appears as an ellipse under oblique viewing and lacks explicit corners, making it infeasible to solve homography via four corner correspondences. A naive approach is to fit the ellipse in the 2D pixel domain and forcibly warp it into a circle; however, this ignores the underlying 3D geometry of perspective imaging and cannot guarantee scale consistency in the metric domain.

We instead revert to 3D geometry: we first recover the tabletop plane and the true circle model using depth, and then construct correspondences from the "planar circle" to the "canonical canvas circle" to estimate the homography. Specifically, the pipeline consists of three stages: **robust contour recovery** (extract stable table-edge pixels from noisy masks), **plane reconstruction and circle fitting** (project edge points onto the 3D plane and fit the true circle), and **canonical canvas mapping** (estimate homography from correspondences between the planar circle and the canvas circle). This strategy ensures scale consistency in the metric domain is grounded in 3D geometric constraints rather than a purely 2D pixel warp. The pipeline is shown in Fig. 4.

a) Stage I: robust contour recovery: We extract the outer contour $\mathcal{C}$ from the mask M, which typically contains gaps and burrs due to segmentation noise. We apply RANSAC-based ellipse fitting [12] to obtain ellipse parameters

$$\mathcal{E}:\ (\mathbf{x}-\mathbf{c}_e)^\top R(\theta)^\top \begin{bmatrix} \frac{1}{a^2} & 0 \\ 0 & \frac{1}{b^2} \end{bmatrix} R(\theta)(\mathbf{x}-\mathbf{c}_e) = 1, \tag{5}$$

and uniformly sample the ellipse at fixed angular steps to obtain a smooth boundary pixel set $\{\mathbf{p}_i\}$, which serves as input to the subsequent 3D projection.

Since depth sensors frequently produce voids and discontinuities near the table edge, directly using the depth at ellipse-sampled pixels may be invalid. For each sampled point $\mathbf{p}_i$, using the ellipse center $\mathbf{c}_e$ as reference, we perform a limited radial inward search to find a corrected point $\tilde{\mathbf{p}}_i$ whose depth is valid and whose corresponding 3D point lies within a threshold distance to the tabletop plane. This step converts the "pixel

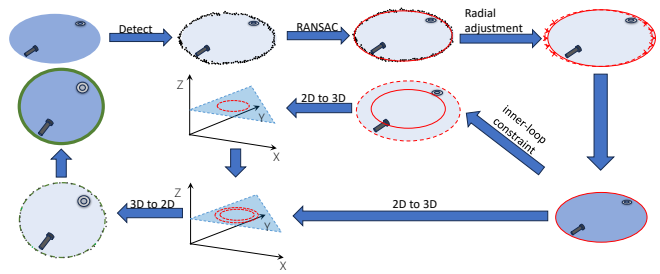

Fig. 4. Round-table metric rectification pipeline. We first obtain robust table-edge pixel points via RANSAC-based ellipse fitting, then recover the tabletop plane from depth and fit the true circle in the plane, and finally estimate the homography from correspondences between the planar circle and the canonical canvas circle.

contour" into "depth-consistent table-edge points," providing reliable input for plane and circle fitting.

b) Stage II: plane reconstruction and circle fitting: We sample a 3D point set $\{\mathbf{P}_j\}$ from the ellipse interior region and fit the tabletop plane via RANSAC:

$$\pi: \ \mathbf{n}^\top(\mathbf{X}-\boldsymbol{\mu})=0, \quad \|\mathbf{n}\|_2=1. \tag{6}$$

Each table-edge 3D point $\tilde{\mathbf{P}}_i$ is orthogonally projected onto the plane:

$$\hat{\mathbf{P}}_i=\tilde{\mathbf{P}}_i-\big(\mathbf{n}^\top(\tilde{\mathbf{P}}_i-\boldsymbol{\mu})\big)\mathbf{n}. \tag{7}$$

We construct an orthonormal basis $(\mathbf{a}_x,\mathbf{a}_y)$ within the plane and convert 3D points to 2D planar coordinates:

$$\mathbf{x}_i=\begin{bmatrix}(\hat{\mathbf{P}}_i-\boldsymbol{\mu})^\top\mathbf{a}_x\\(\hat{\mathbf{P}}_i-\boldsymbol{\mu})^\top\mathbf{a}_y\end{bmatrix}\in\mathbb{R}^2. \tag{8}$$

We then perform RANSAC circle fitting on $\{\mathbf{x}_i\}$ using the minimal sample set of three points, yielding circle center $\mathbf{c}\in\mathbb{R}^2$ and radius r. The inlier criterion is

$$\mathcal{I}=\Big\{\mathbf{x}_i: \ \big|\|\mathbf{x}_i-\mathbf{c}\|_2-r\big|\le\tau_r\Big\}, \tag{9}$$

and we select the optimal circle model by maximizing the inlier count and secondarily minimizing the residual.

c) Stage III: canonical canvas mapping and homography estimation: Let the canvas center be $\mathbf{c}_{\text{can}}=[N/2,N/2]^\top$ and the canvas radius be $r_{\text{can}}=N/2$. We normalize planar circle points to the canonical canvas circle as

$$\mathbf{y}_i=(\mathbf{x}_i-\mathbf{c})\cdot\frac{r_{\text{can}}}{r}+\mathbf{c}_{\text{can}}. \tag{10}$$

We estimate $H_{\text{img}\to\text{can}}$ from the correspondences between pixel source points $\tilde{\mathbf{p}}_i$ and canvas target points $\mathbf{y}_i$. The conversion factor $s_{\text{mm/px}}$ is given by Eq. (2), where W_{mm} is the true diameter of the round tabletop.

We emphasize that Eq. (10) does not mean forcibly warping an ellipse into a circle in the 2D pixel domain. Instead, we recover the true circular geometry on the reconstructed 3D tabletop plane and then define a canonical mapping from the planar circle to the canvas circle, from which the homography is derived. This design ensures scale consistency in the metric domain is supported by explicit geometric foundations.

C. Part Detection and ROI Refinement

1) Outer-structure OBB detection: We use an oriented bounding box (OBB) detector to output the rotated bounding box for each target. For each instance we obtain

$$\mathcal{B}_{\text{out}}=(c_x,c_y,w,h,\alpha), \tag{11}$$

from which the four corners and the midpoints of two pairs of opposite edges can be recovered. The outer OBB serves three roles in the system: filtering off-table false detections, constructing metrology keypoints, and defining the ROI crop region.

2) ROI-based secondary detection for washer inner holes: The inner hole of a washer is a small-scale structure; detecting it directly on the full image is vulnerable to background clutter and scale variation. For each washer outer OBB, we perform rotation-aligned cropping and run a secondary detector inside the local ROI to obtain the inner-hole OBB $\mathcal{B}_{\text{in}}$, as shown in Fig. 5. Let the ROI be generated from the original image via an affine transform T (rotation, translation, and scaling); then detections in ROI coordinates must be strictly mapped back to the original image:

$$\mathbf{p}=T^{-1}(\mathbf{p}'). \tag{12}$$

ROI refinement is used only to improve inner-hole localization stability; final dimension computation is still unified in the metric domain.

D. Keypoint Construction and Metric-Domain Dimension Computation

This section converts detection outputs into metrological quantities in the metric domain. The process consists of two steps: first, keypoints are extracted from OBBs and projected into the metric domain via the homography; second, dimensions are computed in the metric domain. The projection logic is shown in Fig. 6. The core principle is that dimensions are computed from the *distance between projected endpoints*, rather than directly scaling a pixel-length scalar.

1) Keypoint construction and projection: For an arbitrary OBB, we extract the midpoints of the two long edges $\mathbf{m}_{\text{long},1},\mathbf{m}_{\text{long},2}$ and the two short edges $\mathbf{m}_{\text{short},1},\mathbf{m}_{\text{short},2}$ as the metrology keypoint set $\mathcal{K}$. All points in $\mathcal{K}$ are projected to the metric domain via Eq. (1) to obtain $\hat{\mathcal{K}}$. The physical distance between any two keypoints is computed as

$$\ell_{\text{px}}=\|\mathbf{q}_a-\mathbf{q}_b\|_2, \qquad \ell_{\text{mm}}=\ell_{\text{px}}\cdot s_{\text{mm/px}}. \tag{13}$$

2) Washer diameter estimation: Washers lie flat on the tabletop and satisfy the coplanarity assumption; thus height compensation is not needed. The outer and inner diameters are estimated from the outer OBB and the inner-hole OBB in the metric domain, respectively. To mitigate slight anisotropy caused by detection noise, we use the average of two orthogonal directions:

$$d_{\text{out}}=\frac{1}{2}\Big(\|\hat{\mathbf{m}}^{\text{out}}_{\text{long},1}-\hat{\mathbf{m}}^{\text{out}}_{\text{long},2}\|_2+\|\hat{\mathbf{m}}^{\text{out}}_{\text{short},1}-\hat{\mathbf{m}}^{\text{out}}_{\text{short},2}\|_2\Big)\cdot s_{\text{mm/px}}, \tag{14}$$

$$d_{\text{in}}=\frac{1}{2}\Big(\|\hat{\mathbf{m}}^{\text{in}}_{\text{long},1}-\hat{\mathbf{m}}^{\text{in}}_{\text{long},2}\|_2+\|\hat{\mathbf{m}}^{\text{in}}_{\text{short},1}-\hat{\mathbf{m}}^{\text{in}}_{\text{short},2}\|_2\Big)\cdot s_{\text{mm/px}}. \tag{15}$$

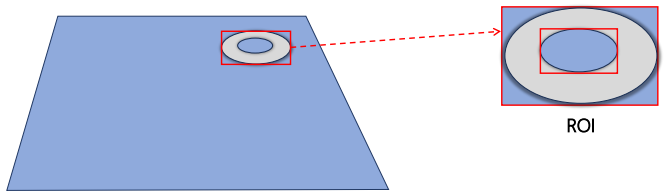

Fig. 5. Two-stage ROI detection. The outer OBB localizes the washer, then a rotation-aligned ROI is cropped and a secondary detector localizes the inner hole. The ROI detection results are mapped back to the original image coordinates via the inverse transform.

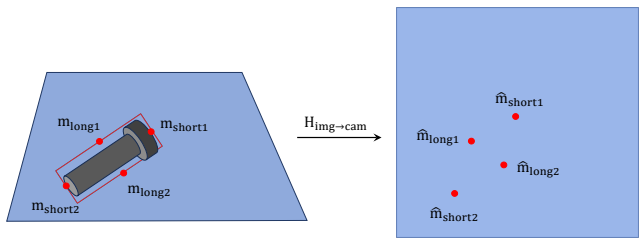

Fig. 6. Keypoint projection and metric computation logic. Midpoints and endpoints from OBBs are first projected to the metric domain via $H_{\text{img}\to\text{can}}$, and Euclidean distances are computed in the metric domain.

3) Planar measurement of bolt length and diameter: Bolt length is first measured in the metric domain under the coplanarity assumption. Let the midpoints of the two short edges of the bolt outer OBB (corresponding to the long-axis endpoints) be $\hat{\mathbf{m}}^{\text{out}}_{\text{short},1}$ and $\hat{\mathbf{m}}^{\text{out}}_{\text{short},2}$. The planar length is

$$L^{(0)} = \|\hat{\mathbf{m}}^{\text{out}}_{\text{short},1} - \hat{\mathbf{m}}^{\text{out}}_{\text{short},2}\|_2 \cdot s_{\text{mm/px}}. \tag{16}$$

Diameter is preferably estimated from the shaft inner structure; if only the outer box is available, we fall back to the short-axis estimate and calibrate the systematic bias using a scale factor γ:

$$d^{(0)} = \gamma \cdot \|\hat{\mathbf{m}}^{\text{out}}_{\text{long},1} - \hat{\mathbf{m}}^{\text{out}}_{\text{long},2}\|_2 \cdot s_{\text{mm/px}}. \tag{17}$$

The above $L^{(0)}$ and $d^{(0)}$ implicitly assume that the measurement points are coplanar with the tabletop. This assumption approximately holds for washers, but bolts have significant height and thus require further compensation, which is the focus of the next subsection.

E. Compensation for Height-Induced Projection Bias of Bolts

When a bolt is placed on the tabletop, its central axis is elevated from the table plane by approximately half of its diameter, i.e., $h \approx d^{(0)}/2$. Under oblique viewing, this height offset causes the bolt center projected onto the tabletop to deviate from its true orthogonal projection, and the offset direction aligns with the projection of the camera viewing ray onto the tabletop plane. The offset magnitude is proportional to the height and the viewing tilt angle. Intuitively, when observing from a tilted angle, the "shadow" of an elevated object shifts along the viewing direction; this is the geometric root cause of the systematic overestimation observed without compensation. In this subsection, we derive interpretable formulas for center-shift compensation and length compensation based on this physical mechanism. The geometry is illustrated in Fig. 7.

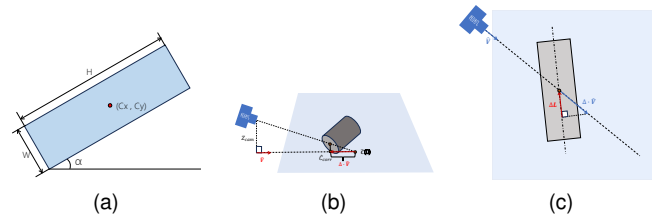

Fig. 7. Illustration of height-induced projection bias for bolts. (a) The bolt center, elevated by height h, induces an in-plane displacement Δ under oblique projection. (b) The in-plane component of the viewing direction and its relation to the tilt angle. (c) Axial decomposition of the length bias.

1) Projection of the viewing direction onto the tabletop plane: Given the intrinsic matrix K and the homography $H_{\text{img}\to\text{can}}$, we can recover the relative pose between the camera and the tabletop. Let the inverse homography from the canvas to the image be $H_{\text{can}\to\text{img}} = H^{-1}_{\text{img}\to\text{can}}$, then

$$\tilde{H} = K^{-1} H_{\text{can}\to\text{img}} \approx [\mathbf{r}_1 \ \mathbf{r}_2 \ \mathbf{t}], \tag{18}$$

and we normalize and orthogonalize $\mathbf{r}_1, \mathbf{r}_2$ to obtain the rotation matrix $R = [\mathbf{r}_1 \ \mathbf{r}_2 \ \mathbf{r}_3]$. The camera optical-axis unit vector $\mathbf{z}_{\text{cam}} = [0, 0, 1]^\top$ expressed in the tabletop coordinate system is

$$\boldsymbol{\ell} = R^\top \mathbf{z}_{\text{cam}} = \begin{bmatrix} \ell_x & \ell_y & \ell_z \end{bmatrix}^\top. \tag{19}$$

The in-plane viewing-direction vector is $\mathbf{v} = [\ell_x, \ell_y]^\top$, with unit vector

$$\hat{\mathbf{v}} = \frac{\mathbf{v}}{\|\mathbf{v}\|_2}. \tag{20}$$

The tilt-angle relation is given by $\tan\theta = \frac{\|\mathbf{v}\|_2}{|\ell_z|}$.

2) Center-shift compensation: The in-plane displacement magnitude induced by the bolt center height $h \approx d^{(0)}/2$ follows a similar-triangle relationship:

$$\Delta = h \tan\theta = h \frac{\|\mathbf{v}\|_2}{|\ell_z|}. \tag{21}$$

The displacement direction is along $\hat{\mathbf{v}}$, and thus the corrected center is

$$\hat{\mathbf{c}}_{\text{corr}} = \hat{\mathbf{c}}^{(0)} - \Delta \cdot \hat{\mathbf{v}}, \tag{22}$$

where $\hat{\mathbf{c}}^{(0)}$ is the initial center position in the metric domain. This compensation directly improves bolt localization accuracy in the tabletop coordinate system.

3) Length compensation: The above center shift also affects length estimation, but the magnitude depends on the angle between the bolt axis and the in-plane viewing direction. Let $\hat{\mathbf{a}}$ be the unit direction of the bolt long axis in the metric domain (determined by the long axis of the outer OBB). The axial component of Δ is

$$\Delta_L = \Delta \cdot |\hat{\mathbf{a}}^\top \hat{\mathbf{v}}|. \tag{23}$$

The compensated length is

$$L = \max(0, \ L^{(0)} - \Delta_L). \tag{24}$$

When the bolt axis is orthogonal to the in-plane viewing direction, $\Delta_L \approx 0$ and the length bias is negligible; when they are parallel, Δ_L reaches its maximum and length overestimation is most significant. This direction-dependent mechanism

explains the empirical observation that "length error is larger than diameter error" (see Section IV-B).

In practice, if the length estimation already meets the required accuracy, one may enable only the center compensation in Eq. (22) to improve localization consistency; if a clear viewpoint-dependent length drift is observed, the length compensation in Eq. (24) can be enabled as well.

F. Algorithm Pseudocode

For reproducibility, we provide pseudocode for the overall system pipeline (Algorithm 1) and the detailed procedure for round-table metric rectification (Algorithm 2). The pseudocode focuses on module-level connections and key conditional branches; algorithmic details of each module have been described in the preceding subsections.

Algorithm 1 Detection-to-Metrology Pipeline for Bolts and Washers

Require: RGB image I, optional depth D, camera intrinsics K, tabletop real size W_{mm}, detector $\mathcal{F}$
Ensure: For each instance: metric center (x, y) and dimensions (bolt L, d; washer $d_{\text{out}}, d_{\text{in}}$)

Obtain tabletop mask $M \leftarrow \text{SegmentTabletop}(I)$
Determine tabletop type type $\in \{\text{square}, \text{round}\}$ from M
if type $==$ square **then**
  $(H_{\mathrm{img}\to\mathrm{can}}, s_{\mathrm{mm/px}}) \leftarrow \text{SquareRectify}(M, W_{\mathrm{mm}})$
else
  $(H_{\mathrm{img}\to\mathrm{can}}, s_{\mathrm{mm/px}}) \leftarrow \text{RoundRectify}(M, D, K, W_{\mathrm{mm}})$
end if
Detect outer OBBs $\{\mathcal{B}^k_{\text{out}}\} \leftarrow \mathcal{F}(I)$
for all instance k **do**
  Construct keypoints $\mathcal{K}^k \leftarrow \text{KeypointsFromOBB}(\mathcal{B}^k_{\text{out}})$
  if instance k is washer **then**
    $\mathcal{B}^k_{\text{in}} \leftarrow \text{ROIRefine}(I, \mathcal{B}^k_{\text{out}})$
    $\mathcal{K}^k \leftarrow \mathcal{K}^k \cup \text{KeypointsFromOBB}(\mathcal{B}^k_{\text{in}})$
  end if
  $\hat{\mathcal{K}}^k \leftarrow H_{\mathrm{img}\to\mathrm{can}}(\mathcal{K}^k)$
  if instance k is bolt **then**
    $(L^{(0)}, d^{(0)}) \leftarrow \text{BoltMeasure}(\hat{\mathcal{K}}^k, s_{\mathrm{mm/px}})$
    Apply height compensation using Eqs. (21)–(24) (optional)
    Output (x, y, L, d)
  else
    $(d_{\text{out}}, d_{\text{in}}) \leftarrow \text{WasherMeasure}(\hat{\mathcal{K}}^k, s_{\mathrm{mm/px}})$
    Output $(x, y, d_{\text{out}}, d_{\text{in}})$
  end if
end for

Algorithm 2 Round Table Metric Rectification via Contour–Plane–Circle Calibration

Require: Tabletop mask M, depth D, intrinsics K, real diameter W_{mm}
Ensure: $H_{\mathrm{img}\to\mathrm{can}}$, $s_{\mathrm{mm/px}}$

$\mathcal{C} \leftarrow \text{FindContour}(M)$
Fit ellipse $\mathcal{E}$ by RANSAC to obtain center $\mathbf{c}_e$, axes (a, b), and angle θ
$\{\mathbf{p}_i\} \leftarrow \text{SampleEllipse}(\mathcal{E})$
for all $\mathbf{p}_i$ **do**
  Radial inward search from $\mathbf{p}_i$ toward $\mathbf{c}_e$ to find valid-depth point $\tilde{\mathbf{p}}_i$
end for
$\{\mathbf{P}_j\} \leftarrow \text{BackProject}(D, K)$
Fit tabletop plane π by RANSAC to obtain $(\mathbf{n}, \boldsymbol{\mu})$
for all $\tilde{\mathbf{p}}_i$ **do**
  Back-project to 3D point $\tilde{\mathbf{P}}_i$
  $\hat{\mathbf{P}}_i \leftarrow \tilde{\mathbf{P}}_i - (\mathbf{n}^\top(\tilde{\mathbf{P}}_i - \boldsymbol{\mu}))\mathbf{n}$
  Convert to plane 2D coordinate $\mathbf{x}_i$ using basis $(\mathbf{a}_x, \mathbf{a}_y)$
end for
Fit circle $\mathcal{C}(\mathbf{c}, r)$ on $\{\mathbf{x}_i\}$ via 3-point RANSAC
Build canvas center $\mathbf{c}_{\text{can}} = [N/2, N/2]^\top$ and radius $r_{\text{can}} = N/2$
Obtain correspondences $\tilde{\mathbf{p}}_i \leftrightarrow \mathbf{y}_i$ via Eq. (10)
Estimate $H_{\mathrm{img}\to\mathrm{can}}$ from the correspondences
$s_{\mathrm{mm/px}} \leftarrow \dfrac{W_{\mathrm{mm}}}{N}$
return $H_{\mathrm{img}\to\mathrm{can}}, s_{\mathrm{mm/px}}$

IV. Experiments

We first evaluate detection and measurement performance on RGB images and point-cloud data acquired by a commercial AstraProPlus camera under both square-table and round-table conditions. To demonstrate the accuracy of the proposed algorithm, we conduct experiments using (i) manually annotated ground-truth RGB bounding boxes to estimate bolt and washer dimensions and (ii) model-predicted OBB boxes, and we further perform ablation studies with and without height compensation to verify the importance of height compensation for bolts.

To comprehensively evaluate the proposed height-compensated dimension estimation framework, we perform systematic experiments on bolt and washer measurement tasks. The experiments use RGB images and point-cloud data collected by an Orbbec AstraProPlus commercial RGB-D camera. We consider two representative industrial scenarios: a square worktable and a round worktable, which provide different geometric constraints and camera viewpoints.

A. Experimental Setup

Hardware. We use an Orbbec AstraProPlus depth camera rigidly mounted above the worktable. The camera provides synchronized RGB images and depth maps at a resolution of 640×480.

Test objects. The evaluation covers hex bolts with diameters ranging from 6 mm to 16 mm and lengths from 10 mm to 150 mm, as well as various flat washers with outer diameters of 12–24 mm and inner diameters of 3–14 mm.

Evaluation metrics. We adopt five error metrics: mean absolute error (MAE), root mean square error (RMSE), 95th-percentile error (P95), maximum error (Max), and systematic bias (Bias). In addition, we report accuracy at multiple thresholds: Acc@3mm, Acc@5mm, and Acc@8mm, defined as the percentage of samples whose absolute prediction error is within τ mm.

Experimental conditions. To isolate the contribution of each component, we evaluate under four settings:

- **GT box + height compensation**: ground-truth boxes with height compensation
- **GT box + no height compensation**: ground-truth boxes without height compensation
- **Predicted box + height compensation**: predicted OBB boxes with height compensation
- **Predicted box + no height compensation**: predicted boxes without height compensation

B. Bolt Dimension Estimation Results

Tables I and III report the comprehensive bolt estimation results under the square-table and round-table scenarios, respectively. The proposed height compensation method achieves significant improvements across all metrics.

Figure 8 visualizes the MAE comparison across different settings. The key observations are as follows.

Width/diameter estimation. Under the GT-box + height compensation setting, the width MAE is 1.37 mm (square table) and 1.77 mm (round table), with Acc@5mm exceeding 96% in both scenarios. Without height compensation, the MAE increases to 3.79 mm and 3.90 mm, respectively, indicating an approximately $2.5\times$ degradation.

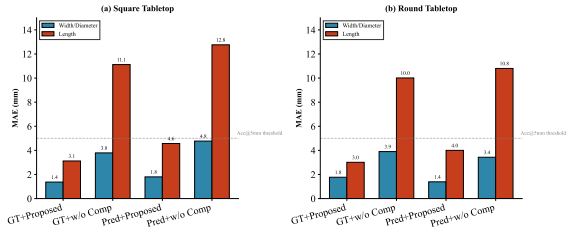

Fig. 8. Mean absolute error comparison of bolt width/diameter and length estimation under four experimental settings. The dashed line indicates the Acc@5mm threshold. (a) Square-table results. (b) Round-table results.

Length estimation. Height compensation has an even more pronounced impact on length estimation. The proposed method achieves MAE of 3.12 mm (square) and 3.01 mm (round), whereas the baseline without compensation yields MAE of 11.12 mm and 10.01 mm—a 3.3× increase. This substantial gap arises because length estimation is more sensitive to perspective distortion along the camera viewing direction.

Figure 9 shows accuracy as a function of the tolerance threshold τ. The proposed method maintains clear advantages at all thresholds. Notably, for length estimation on the square table, GT + height compensation achieves 86.80% accuracy at $\tau = 5$ mm, whereas the baseline achieves only 32.99%, improving by 54 percentage points.

C. Ablation Study: Effectiveness of Height Compensation

To rigorously validate the contribution of height compensation, we conduct controlled ablation experiments with and without this component under the same GT-box setting.

Figure 10 presents the dual-axis ablation results. Height compensation yields consistent improvements:

- **Square table**: width MAE decreases by 64% (3.79 mm → 1.37 mm); length MAE decreases by 72% (11.12 mm → 3.12 mm).
- **Round table**: width MAE decreases by 55% (3.90 mm → 1.77 mm); length MAE decreases by 70% (10.01 mm → 3.01 mm).

The systematic bias analysis in Table I further reveals the underlying mechanism. Without height compensation, the length bias reaches 11.12 mm (square) and 9.94 mm (round), indicating consistent overestimation caused by uncorrected perspective projection. The proposed method reduces the bias to 1.25 mm and 0.69 mm, effectively eliminating the systematic error.

D. Washer Dimension Estimation Results

Tables II and IV summarize the washer diameter estimation performance. Unlike bolts, washers lie flat on the surface and therefore do not require height compensation.

Figure 11 compares washer estimation performance. All configurations achieve Acc@5mm above 90%, demonstrating the robustness of the ellipse-fitting-based method. Notably, inner diameter estimation consistently outperforms outer diameter estimation, with MAE differing by 0.5–0.8 mm. This can be attributed to the sharper edge contrast at the inner boundary, which facilitates more accurate ellipse fitting.

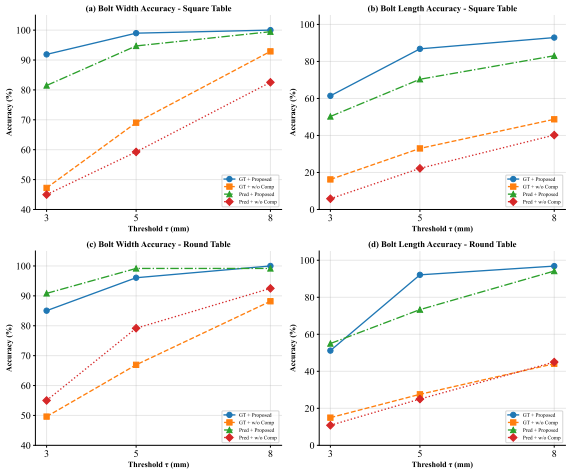

Fig. 9. Accuracy–threshold curves for bolt estimation. (a)(b) Width accuracy on square and round tables. (c)(d) Length accuracy on square and round tables. The proposed method (blue solid line) consistently outperforms the baseline at all thresholds.

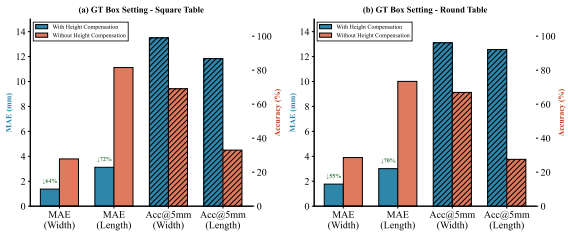

Fig. 10. Ablation study of height compensation under GT-box settings. Left axis: MAE (mm). Right axis: Acc@5mm (%). Green annotations indicate relative MAE reduction. (a) Square table. (b) Round table.

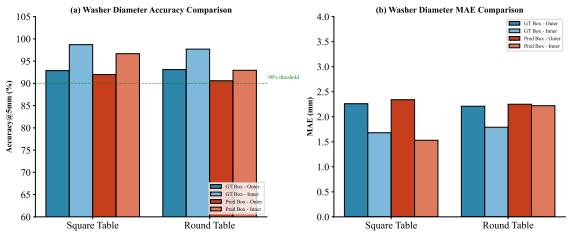

Fig. 11. Washer diameter estimation comparison. (a) Acc@5mm for outer and inner diameters; the dashed line indicates the 90% threshold. (b) MAE comparison across settings and table types.

The performance gap between GT-box and predicted-box settings on washers is small (Acc@5mm differs by less than 2%), indicating that the washer detection module provides reliable bounding boxes and does not significantly affect downstream dimension estimation.

E. Center Localization Accuracy

Accurate center localization is crucial for robotic grasping applications. We evaluate localization accuracy along the X and Y axes separately because camera geometry induces different error characteristics.

Figure 12(a) reveals a notable finding: Y-axis correction is critical for precise localization. Under the GT-box setting, Y-axis Acc@5mm improves dramatically from 10.36% (uncorrected) to 93.26% (corrected), a gain of 83 percentage points.

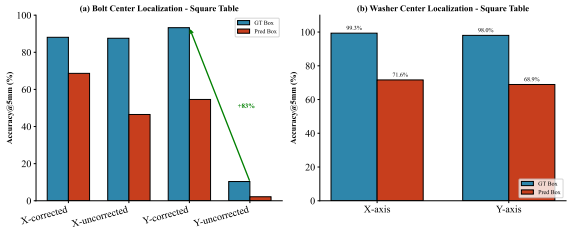

Fig. 12. Center localization accuracy on the square table. (a) Bolt center localization with/without coordinate correction. Green arrows indicate an 83% improvement in Y-axis accuracy after correction. (b) Washer center localization accuracy.

This improvement stems from the camera pitch tilt, which makes perspective distortion more pronounced along the Y direction and induces large systematic localization bias without correction.

For washer center localization (Fig. 12(b)), the GT-box setting achieves 99.3% (X-axis) and 98.0% (Y-axis) at the 5 mm threshold, meeting the accuracy requirements for automated assembly tasks.

F. Radial Distance Accuracy on the Round Table

For the round-table scenario, we additionally evaluate radial distance estimation, i.e., the distance from each fastener to the tabletop center.

Figure 13(a) shows that height compensation is also important for radial distance estimation. With compensation, GT + compensation achieves MAE of 3.76 mm and Acc@5mm of 96.85%. Without compensation, MAE increases to 10.20 mm and Acc@5mm drops to 25.20%, representing an approximately $4\times$ degradation.

Figure 13(b) compares washer radial distance estimation using model-labeled versus manually labeled tabletop centers. The two results are comparable (MAE: 3.86 mm vs. 4.13 mm), validating the reliability of our automatic center detection algorithm, which can replace manual calibration in practical deployment.

G. Comprehensive Performance Analysis

To provide a global view of performance, we present summary visualizations that consolidate results across all scenarios and settings.

Figure 14 visualizes Acc@3mm, Acc@5mm, and Acc@8mm across all 12 configurations (4 bolt settings $\times$ 2 table types + 2 washer settings $\times$ 2 table types). The following patterns can be observed:

- **Advantage of the proposed method:** all "+ compensation" rows are dominated by green tones (Acc@5mm $>$ 90%), whereas "no compensation" rows show yellow/red tones, especially under stricter thresholds.
- **Stability of washers:** washer estimation remains consistently high across GT and predicted boxes, indicating robust edge localization for washers.
- **Cross-scenario consistency:** improvements are consistent between square and round tables, demonstrating the generalization capability of the proposed framework.

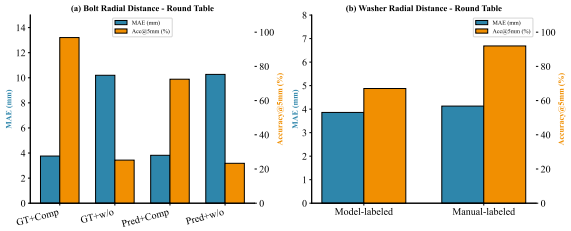

Fig. 13. Radial distance estimation accuracy on the round table. Left axis: MAE (mm). Right axis: Acc@5mm (%). (a) Bolt radial distance under four settings. (b) Washer radial distance comparing model-labeled and manually labeled center references.

Fig. 14. Performance heatmap of width/outer-diameter estimation accuracy under 12 experimental configurations. Color scale: green indicates high accuracy ($>$90%), yellow indicates medium accuracy (50–90%), and red indicates low accuracy ($<$50%).

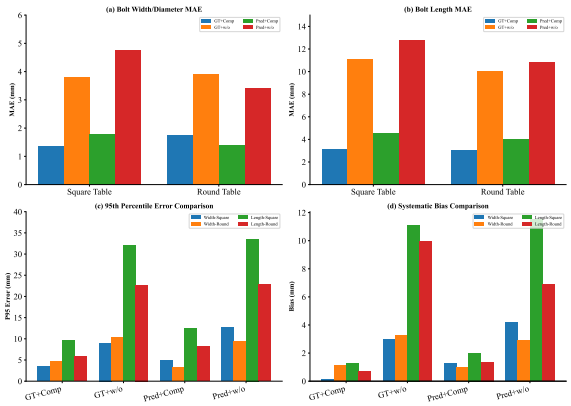

Fig. 15. Comparison of comprehensive error metrics. (a) Bolt width/diameter MAE. (b) Bolt length MAE. (c) 95th-percentile error distribution. (d) Bias analysis—values near zero indicate unbiased estimation.

Figure 15 provides a detailed error breakdown. The P95 analysis (Fig. 15(c)) shows that without height compensation, the 95th-percentile length error can reach 32 mm, which is unacceptable for precision assembly. The bias analysis (Fig. 15(d)) confirms that the proposed method effectively removes systematic bias: GT + compensation achieves near-zero bias ($<$1.5 mm) across measurements, while the baseline exhibits significant positive bias (up to 11 mm) due to perspective-induced overestimation.

H. End-to-End Performance: GT Boxes vs. Predicted Boxes

For practical deployment, a key question is how detection noise affects overall system performance. We analyze this by comparing GT-box (upper bound) and predicted-box (deployment) settings.

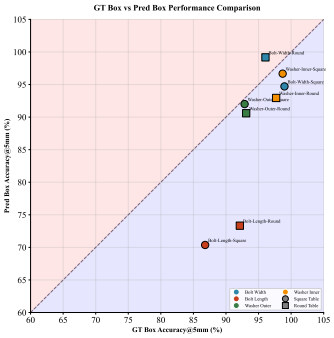

Fig. 16. Scatter plot comparing Acc@5mm under GT-box and predicted-box settings. Points above the diagonal indicate predicted boxes outperform GT boxes; points below indicate GT advantage. The blue shaded area marks the typical GT-advantage region.

Figure 16 presents the scatter analysis, where each point corresponds to a specific measurement type on a specific tabletop. The main observations include:

- Most points cluster in the blue region (GT $>$ predicted), confirming the expected performance hierarchy.
- The performance gap is smallest for bolt width and washer inner diameter ($<$5%), where edge localization is more reliable.
- The gap is largest for bolt length (GT: 87% $\rightarrow$ predicted: 70%), since length estimation is more sensitive to bounding-box alignment errors.
- Importantly, all predicted-box results maintain Acc@5mm above 70%, indicating practical utility even with imperfect detection.

I. Summary of Improvements

Figure 17 provides a comprehensive summary using radar charts over six performance dimensions: width Acc@5mm, length Acc@5mm, and inversely scaled MAE and P95 for both width and length. The blue region (with compensation) consistently encloses the red region (without), indicating that:

1) Height compensation improves *all* measurement aspects rather than only selected metrics.
2) The improvement pattern is consistent between square and round tables.
3) The most significant gains appear in accuracy metrics (Acc@5mm), where the blue boundary clearly exceeds the red boundary.

J. Discussion

The experimental results validate three core claims:

(1) Height compensation is indispensable. Across all scenarios, the height compensation module reduces MAE by 55–72% and improves Acc@5mm by up to 54 percentage points. Ablation confirms that the improvement stems from correcting systematic perspective distortion rather than reducing random noise.

(2) The framework generalizes across scenarios. Consistent performance on square and round tables demonstrates that the geometric correction principles are not specific to a particular scenario. The method adapts to different camera–object configurations without scenario-specific tuning.

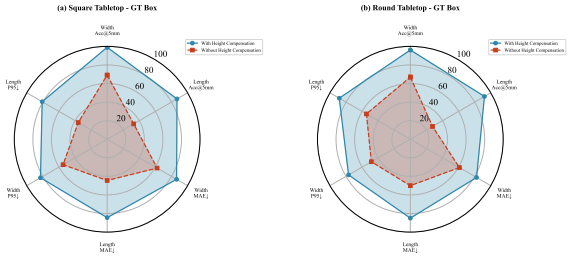

Fig. 17. Radar chart summarizing multi-dimensional improvements brought by height compensation under GT-box settings. Blue: with compensation; red: without. Larger area indicates better overall performance. (a) Square table. (b) Round table.

(3) End-to-end deployment is feasible. Although predicted-box results are 5–15% lower than GT-box results in accuracy, the absolute performance (Acc@5mm $>$ 70% for most metrics) satisfies industrial automation requirements. Robustness to detection noise ensures practical applicability.

V. Conclusion

This paper proposes a visual dimension estimation framework for bolts and washers in oblique-view industrial tabletop scenarios. The core idea is to unify geometric computation in a tabletop metric domain: by establishing a stable homography mapping and scale reference from the tabletop planar structure, and performing robust geometric fitting in the metric domain under boundary constraints from high-quality RGB masks, the framework avoids the disruption of point-cloud-endpoint measurements caused by depth voids from metallic specular reflections. For square and round tabletops, we design homography estimation methods based on quadrilateral regularization and a "contour–plane–circle" strategy, respectively. For the systematic perspective-projection bias introduced by bolt non-coplanarity, we propose an interpretable height compensation mechanism grounded in imaging geometry.

Systematic experiments validate the effectiveness and practicality of the proposed framework. Under GT-box settings (excluding detection error), key dimension estimation for bolts and washers generally exceeds 90% under Acc@5mm, demonstrating stable millimeter-level accuracy of the metrology algorithm itself. Ablation experiments show that height compensation reduces bolt length MAE by 70–72% and compresses systematic bias from approximately 10 mm to within 1.5 mm, effectively eliminating consistent overestimation caused by uncorrected perspective projection. Under end-to-end predicted-box settings, with RANSAC-based robust processing, dimension estimation remains stably above 80% at Acc@8mm, demonstrating tolerance to detection noise. Localization evaluation shows that center position errors for bolts and washers are overall controlled within 10 mm, and Y-axis correction yields an 83 percentage-point improvement, satisfying tolerance requirements for most industrial grasping and assembly tasks. The performance remains highly consistent between square and round tabletop scenarios, confirming cross-scenario generalization of the geometric rectification principles.

The project repository is available at https://github.com/lqx943576099/Visual-Dimension-Estimation.

A. Limitations and Future Work

Although the proposed framework achieves satisfactory results in controlled industrial scenarios, several limitations warrant further investigation.

First, the current homography estimation relies on the known true tabletop size as a metric reference. Future work may explore automatic absolute-scale recovery using local point-cloud geometric constraints or multi-view observations, thereby reducing reliance on prior calibration information and enabling more flexible deployment.

Second, our experiments are conducted in relatively clean tabletop environments with limited occlusion and background clutter. In real production lines, part stacking, oil contamination, and drastic illumination changes may pose greater challenges for detection and boundary extraction. Introducing stronger instance segmentation models and more comprehensive data augmentation strategies to improve robustness under complex conditions is an important direction.

Third, the height compensation currently assumes a simplified cylindrical geometry ($h \approx d/2$), which may be biased for irregular fasteners or non-standard placement poses. Incorporating deep-learning-based height prediction or finer 3D shape priors could further improve compensation accuracy.

Finally, the millimeter-level dimensions and planar coordinates produced by the framework can naturally serve as geometric priors for downstream robotic operations. Closing the loop by integrating metrology outputs with grasp planning and assembly sequence decision modules to build a complete perception-to-action industrial intelligent system is a valuable future direction.

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APPENDIX A
DETAILED EXPERIMENTAL TABLES

TABLE I
BOLT LENGTH AND WIDTH (DIAMETER) ESTIMATION RESULTS UNDER THE SQUARE-TABLE SCENARIO.

Setting	Width / Diameter error (W)					Width accuracy (W)			Length error (L)					Length accuracy (L)		
	$\mathbf{MAE}_W$	$\mathbf{RMSE}_W$	$\mathbf{P95}_W$	$\mathbf{Max}_W$	$\mathbf{Bias}_W$	$\mathbf{Acc}_W$**@3**	$\mathbf{Acc}_W$**@5**	$\mathbf{Acc}_W$**@8**	$\mathbf{MAE}_L$	$\mathbf{RMSE}_L$	$\mathbf{P95}_L$	$\mathbf{Max}_L$	$\mathbf{Bias}_L$	$\mathbf{Acc}_L$**@3**	$\mathbf{Acc}_L$**@5**	$\mathbf{Acc}_L$**@8**
GT box + proposed estimator	1.37	1.76	3.42	7.5	0.14	91.88%	98.98%	100%	3.12	4.54	9.6	19.2	1.25	61.42%	86.80%	92.89%
GT box + w/o height compensation	3.79	4.8	9.04	17.1	2.95	47.21%	69.04%	92.89%	11.12	14.54	32.02	51.6	11.12	16.24%	32.99%	48.73%
Pred box + proposed estimator	1.8	2.41	4.86	8.7	1.24	81.48%	94.71%	99.47%	4.57	7.2	12.5	48.3	2	50.26%	70.37%	83.07%
Pred box + w/o height compensation	4.77	6.16	12.74	16.4	4.17	44.97%	59.26%	82.54%	12.76	16.26	33.58	57.3	11.49	5.82%	22.22%	40.21%

Note: All errors are in mm. For width/diameter W (i.e., bolt diameter d), MAE_W/RMSE_W/$\mathrm{P95}_W$/Max_W/Bias_W are computed from $e_i^W = \hat{d}_i - d_i$. For length L, MAE_L/RMSE_L/$\mathrm{P95}_L$/Max_L/Bias_L are computed from $e_i^L = \hat{L}_i - L_i$. $\mathrm{Acc}_W@\tau = \frac{1}{N}\sum_{i=1}^{N}\mathbb{I}(|\hat{d}_i - d_i| \le \tau)$ and $\mathrm{Acc}_L@\tau = \frac{1}{N}\sum_{i=1}^{N}\mathbb{I}(|\hat{L}_i - L_i| \le \tau)$.

TABLE II
WASHER DIAMETER ESTIMATION RESULTS (OUTER/INNER) UNDER THE SQUARE-TABLE SCENARIO.

Setting	Outer error (mm)					Inner error (mm)					$\mathbf{Acc}_{\mathbf{out}}$			$\mathbf{Acc}_{\mathbf{in}}$		
	MAE	**RMSE**	**P95**	**Max**	**Bias**	**MAE**	**RMSE**	**P95**	**Max**	**Bias**	**@3**	**@5**	**@8**	**@3**	**@5**	**@8**
GT box + proposed estimator	2.26	2.79	5.37	7.9	2.07	1.68	2.87	3.3	27.3	0.91	71.43%	92.86%	100.00%	88.31%	98.70%	99.35%
Pred box + proposed estimator	2.34	3.16	6.24	13.5	1.84	1.53	2.35	4.10	16.9	−1.05	72.67%	92.00%	98.00%	89.33%	96.67%	99.33%

Note: All errors are in mm. For outer/inner diameter $d_{\text{out}}, d_{\text{in}}$, $\mathrm{MAE} = \frac{1}{N}\sum_i |\hat{d}_i - d_i|$, $\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_i(\hat{d}_i - d_i)^2}$, P95 is the 95th percentile of $|\hat{d} - d|$, $\mathrm{Max} = \max_i |\hat{d}_i - d_i|$, $\mathrm{Bias} = \frac{1}{N}\sum_i(\hat{d}_i - d_i)$. $\mathrm{Acc}@\tau$ denotes $\frac{1}{N}\sum_i \mathbb{I}(|\hat{d}_i - d_i| \le \tau)$.

TABLE III
BOLT LENGTH AND WIDTH (DIAMETER) ESTIMATION RESULTS UNDER THE ROUND-TABLE SCENARIO.

Setting	Width/diameter d error (mm)					$\mathbf{Acc}_W$			Length L error (mm)					$\mathbf{Acc}_L$		
	MAE	**RMSE**	**P95**	**Max**	**Bias**	**@3**	**@5**	**@8**	**MAE**	**RMSE**	**P95**	**Max**	**Bias**	**@3**	**@5**	**@8**
GT box + proposed estimator	1.77	2.19	4.64	6.2	1.10	85.04%	96.06%	100.00%	3.01	3.98	5.81	20.4	0.69	51.18%	92.13%	96.85%
GT box + w/o height compensation	3.90	5.04	10.31	14.0	3.24	49.61%	66.93%	88.19%	10.01	11.95	22.72	32.5	9.94	14.96%	27.56%	44.09%
Pred box + proposed estimator	1.39	2.06	3.20	13.7	0.99	90.83%	99.17%	99.17%	4.00	7.58	8.15	68.2	1.37	55.00%	73.33%	94.17%
Pred box + w/o height compensation	3.43	4.74	9.32	26.1	2.92	55.00%	79.17%	92.50%	10.8	13.89	22.88	77.2	6.9	10.83%	25.00%	45.00%

Note: All errors are in mm. For width/diameter d and length L, $\mathrm{MAE} = \frac{1}{N}\sum_i |\hat{v}_i - v_i|$, $\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_i(\hat{v}_i - v_i)^2}$, P95 is the 95th percentile of $|\hat{v} - v|$, $\mathrm{Max} = \max_i |\hat{v}_i - v_i|$, $\mathrm{Bias} = \frac{1}{N}\sum_i(\hat{v}_i - v_i)$, where $v \in \{d, L\}$. $\mathrm{Acc}_W@\tau = \frac{1}{N}\sum_i \mathbb{I}(|\hat{d}_i - d_i| \le \tau)$ and $\mathrm{Acc}_L@\tau = \frac{1}{N}\sum_i \mathbb{I}(|\hat{L}_i - L_i| \le \tau)$.

TABLE IV
WASHER DIAMETER ESTIMATION RESULTS (OUTER/INNER) UNDER THE ROUND-TABLE SCENARIO.

	Outer diameter d_{out}								Inner diameter d_{in}							
Setting	**MAE**	**RMSE**	**P95**	**Max**	**Bias**	**Acc@3**	**Acc@5**	**Acc@8**	**MAE**	**RMSE**	**P95**	**Max**	**Bias**	**Acc@3**	**Acc@5**	**Acc@8**
GT box + proposed estimator	2.21	2.77	5.17	6.7	1.41	75.86%	93.10%	100.00%	1.79	3.72	3.6	27.6	0.59	89.66%	97.70%	97.70%
Pred box + proposed estimator	2.25	3.04	5.5	13	1.38	70.59%	90.59%	98.82%	2.22	3.01	5.26	13.9	−1.95	80.00%	92.94%	97.65%

Note: All errors are in mm.

TABLE V
AXIS-WISE BOLT CENTER LOCALIZATION ACCURACY UNDER THE SQUARE-TABLE SCENARIO (ASTRAPROPLUS).

Setting	**X error (mm)**					**Y error (mm)**					**Acc$_x$ (%)**			**Acc$_y$ (%)**		
	MAE	**RMSE**	**P95**	**Max**	**Bias**	**MAE**	**RMSE**	**P95**	**Max**	**Bias**	**@3**	**@5**	**@8**	**@3**	**@5**	**@8**
GT box + corrected	2.65	3.76	7.20	19.7	0.22	5.60	11.55	8.20	101.4	3.68	70.98	88.08	96.89	71.55	93.26	93.78
GT box + uncorrected	2.76	3.93	7.08	19.5	0.95	13.00	17.25	24.88	91.1	−12.06	69.95	87.56	95.34	1.55	10.36	31.09
Pred box + corrected	4.67	6.60	14.10	22.1	0.07	6.66	12.28	14.56	92.8	0.29	53.51	68.65	77.84	32.43	54.59	75.68
Pred box + uncorrected	7.06	8.82	17.70	24.1	−0.46	14.82	19.12	25.80	103.5	4.66	24.32	46.49	70.27	0.00	2.16	17.30

Note: Errors are in mm. For axis $a \in \{x, y\}$, let $e_a = \hat{a} - a$ be the signed error. MAE= $\frac{1}{N}\sum_i |e_a^{(i)}|$, RMSE= $\sqrt{\frac{1}{N}\sum_i (e_a^{(i)})^2}$, P95 is the 95th percentile of $|e_a|$, Max= $\max_i |e_a^{(i)}|$, Bias= $\frac{1}{N}\sum_i e_a^{(i)}$, and Acc@$\tau = \frac{1}{N}\sum_i \mathbb{I}(|e_a^{(i)}| \le \tau)$.

TABLE VI
AXIS-WISE WASHER CENTER LOCALIZATION ACCURACY UNDER THE SQUARE-TABLE SCENARIO (ASTRAPROPLUS).

Setting	**X error (mm)**					**Y error (mm)**					**Acc$_x$ (%)**			**Acc$_y$ (%)**		
	MAE	**RMSE**	**P95**	**Max**	**Bias**	**MAE**	**RMSE**	**P95**	**Max**	**Bias**	**@3**	**@5**	**@8**	**@3**	**@5**	**@8**
GT box + proposed estimator (rectified)	1.20	1.57	3.40	5.50	0.58	3.23	9.88	4.70	88.60	−0.65	92.11	99.34	100.00	69.74	98.03	98.03
Pred box + proposed estimator (rectified)	4.17	5.67	11.99	21.90	0.85	5.34	11.61	15.97	86.30	3.91	53.38	71.62	82.43	54.05	68.92	83.78

Note: Let $e_x = \hat{x} - x$ and $e_y = \hat{y} - y$ (in mm). For each axis, MAE= $\frac{1}{N}\sum |e|$, RMSE= $\sqrt{\frac{1}{N}\sum e^2}$, P95 is the 95th percentile of $|e|$, Max= $\max |e|$, and Bias= $\frac{1}{N}\sum e$. Acc$_x$@$\tau = \frac{1}{N}\sum \mathbb{I}(|e_x| \le \tau)$ and Acc$_y$@$\tau = \frac{1}{N}\sum \mathbb{I}(|e_y| \le \tau)$, reported in %.

TABLE VII
RADIAL DISTANCE ACCURACY OF BOLTS TO THE ROUND-TABLE CENTER UNDER THE ROUND-TABLE SCENARIO (ASTRAPROPLUS).

Setting	**MAE$_r$**	**RMSE$_r$**	**P95$_r$**	**Max$_r$**	**Bias$_r$**	**Acc@3$_r$**	**Acc@5$_r$**	**Acc@8$_r$**
GT center + height-comp	3.76	4.47	4.97	26.1	−1.24	20.47%	96.85%	98.43%
GT center + w/o height-comp	10.20	12.00	20.50	30.4	2.68	11.81%	25.20%	40.16%
Pred center (ellipse-fit) + height-comp	3.82	5.29	10.72	19.7	−1.00	51.67%	72.50%	87.50%
Pred center (ellipse-fit) + w/o height-comp	10.27	11.87	20.59	29.6	0.58	7.50%	23.33%	35.83%

Note: r is the distance (mm) from the object center to the estimated circular tabletop center. Let $e_r^{(i)} = \hat{r}_i - r_i$ be the signed error. MAE$_r = \frac{1}{N}\sum_i |e_r^{(i)}|$, RMSE$_r = \sqrt{\frac{1}{N}\sum_i (e_r^{(i)})^2}$, P95$_r$ is the 95th percentile of $|e_r|$, Max$_r = \max_i |e_r^{(i)}|$, Bias$_r = \frac{1}{N}\sum_i e_r^{(i)}$, and Acc@$\tau_r = \frac{1}{N}\sum_i \mathbb{I}(|e_r^{(i)}| \le \tau)$.

TABLE VIII
RADIAL DISTANCE ACCURACY OF WASHERS TO THE ROUND-TABLE CENTER UNDER THE ROUND-TABLE SCENARIO (ASTRAPROPLUS).

Setting	**MAE$_r$**	**RMSE$_r$**	**P95$_r$**	**Max$_r$**	**Bias$_r$**	**Acc@3$_r$**	**Acc@5$_r$**	**Acc@8$_r$**
Model-labeled center (ellipse fitting)	3.86	4.76	8.58	14.0	−1.46	50.59%	67.06%	92.94%
Manual-labeled center (ellipse fitting)	4.13	5.00	8.44	21.2	−0.90	14.94%	91.95%	94.25%

Note: r denotes the distance (mm) from the washer center to the estimated circular tabletop center. Let $e_r^{(i)} = \hat{r}_i - r_i$ be the signed error. MAE$_r = \frac{1}{N}\sum_i |e_r^{(i)}|$, RMSE$_r = \sqrt{\frac{1}{N}\sum_i (e_r^{(i)})^2}$, P95$_r$ is the 95th percentile of $|e_r|$, Max$_r = \max_i |e_r^{(i)}|$, Bias$_r = \frac{1}{N}\sum_i e_r^{(i)}$, and Acc@$\tau_r = \frac{1}{N}\sum_i \mathbb{I}(|e_r^{(i)}| \le \tau)$ (reported in %).